

Rajalakshmi Engineering College

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NeoColab_REC_CS23231_DATA STRUCTURES

REC_DS using C_Week 2_COD_Question 4

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Ravi is developing a student registration system for a college. To efficiently store and manage the student IDs, he decides to implement a doubly linked list where each node represents a student's ID.

In this system, each student's ID is stored sequentially, and the system needs to display all registered student IDs in the order they were entered.

Implement a program that creates a doubly linked list, inserts student IDs, and displays them in the same order.

Input Format

The first line contains an integer N the number of student IDs.

The second line contains N space-separated integers representing the student IDs.

Output Format

The output should display the single line containing N space-separated integers representing the student IDs stored in the doubly linked list.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

10 20 30 40 50

Output: 10 20 30 40 50

Answer

```
#include<stdio.h>
#include<stdlib.h>
typedef struct Node{
    int data;
    struct Node*next;
    struct Node*prev;
}Node;

void insertEnd(Node**head,int data){
    Node*newNode=(Node*)malloc(sizeof(Node));
    newNode->data=data;
    newNode->next=NULL;

    if(*head==NULL){
        newNode->prev=NULL;
        *head=newNode;
        return;
    }
    Node*temp=*head;
    while(temp->next!=NULL){
        temp=temp->next;
    }
    temp->next=newNode;
    newNode->prev=temp;
```

```
}
```

```
void printList(Node*head){  
    Node*temp=head;  
    while(temp!=NULL){  
        printf("%d",temp->data);  
        temp=temp->next;  
    }  
    printf("\n");  
}
```

```
int main(){  
    int N,id;  
    Node*head=NULL;  
    scanf("%d",&N);  
    for (int i=0;i<N;i++){  
        scanf("%d",&id);  
        insertEnd(&head,id);  
    }  
    printList(head);  
    return 0;  
}
```

Status : Correct

Marks : 10/10